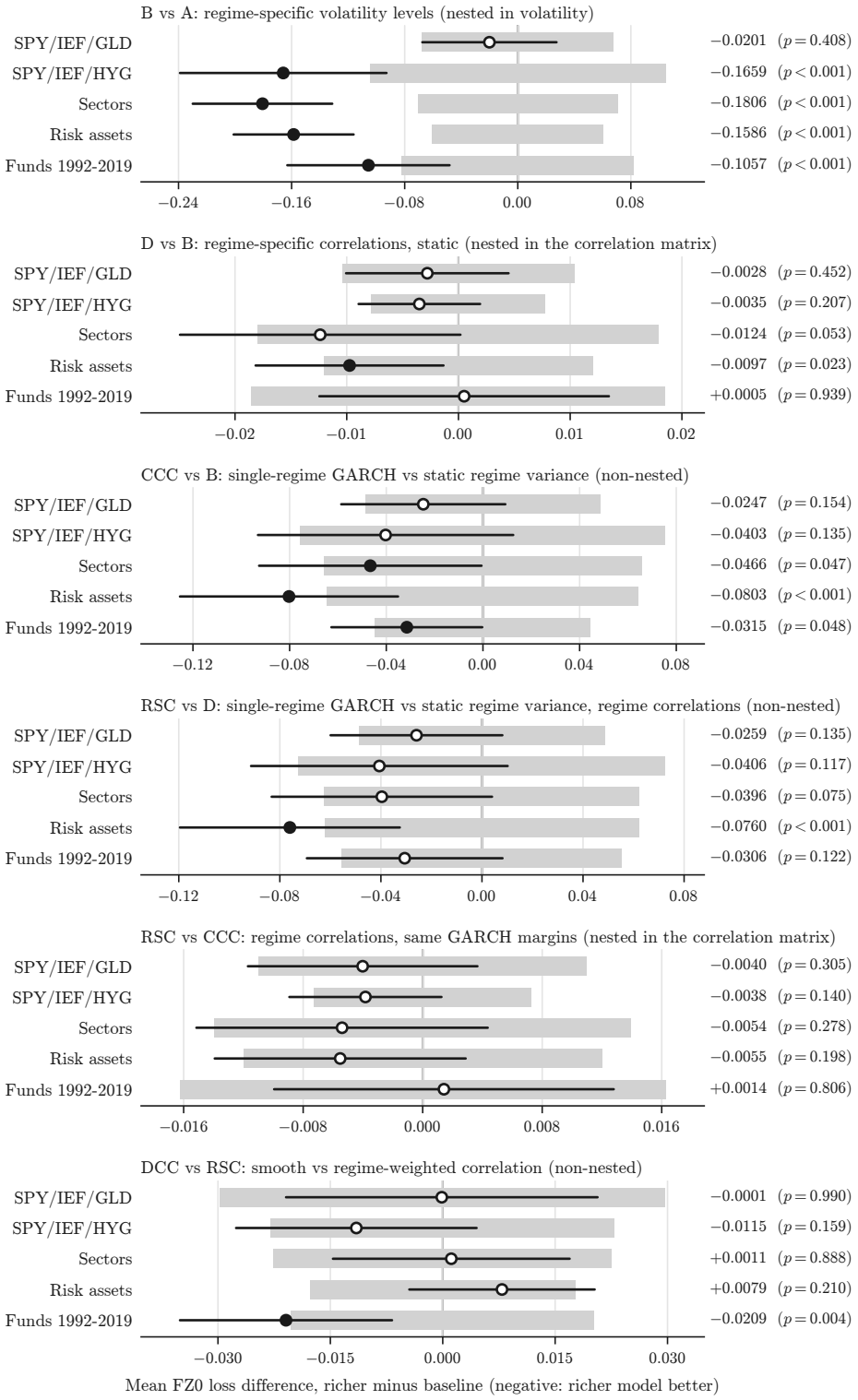
B vs A: regime-specific volatility levels (nested in volatility)
SPY/IEF/GLD
SPY/IEF/HYG
Sectors
Risk assets
Funds 1992-2019
−0.0201 (p = 0.408)
−0.1659 (p < 0.001)
−0.1806 (p < 0.001)
−0.1586 (p < 0.001)
−0.1057 (p < 0.001)
−0.24
−0.16
−0.08
0.00
0.08
D vs B: regime-specific correlations, static (nested in the correlation matrix)
SPY/IEF/GLD
SPY/IEF/HYG
Sectors
Risk assets
Funds 1992-2019
−0.0028 (p = 0.452)
−0.0035 (p = 0.207)
−0.0124 (p = 0.053)
−0.0097 (p = 0.023)
+0.0005 (p = 0.939)
−0.02
−0.01
0.00
0.01
0.02
CCC vs B: single-regime GARCH vs static regime variance (non-nested)
SPY/IEF/GLD
SPY/IEF/HYG
Sectors
Risk assets
Funds 1992-2019
−0.0247 (p = 0.154)
−0.0403 (p = 0.135)
−0.0466 (p = 0.047)
−0.0803 (p < 0.001)
−0.0315 (p = 0.048)
−0.12
−0.08
−0.04
0.00
0.04
0.08
RSC vs D: single-regime GARCH vs static regime variance, regime correlations (non-nested)
SPY/IEF/GLD
SPY/IEF/HYG
Sectors
Risk assets
Funds 1992-2019
−0.0259 (p = 0.135)
−0.0406 (p = 0.117)
−0.0396 (p = 0.075)
−0.0760 (p < 0.001)
−0.0306 (p = 0.122)
−0.12
−0.08
−0.04
0.00
0.04
0.08
RSC vs CCC: regime correlations, same GARCH margins (nested in the correlation matrix)
SPY/IEF/GLD
SPY/IEF/HYG
Sectors
Risk assets
Funds 1992-2019
−0.0040 (p = 0.305)
−0.0038 (p = 0.140)
−0.0054 (p = 0.278)
−0.0055 (p = 0.198)
+0.0014 (p = 0.806)
−0.016
−0.008
0.000
0.008
0.016
DCC vs RSC: smooth vs regime-weighted correlation (non-nested)
SPY/IEF/GLD
SPY/IEF/HYG
Sectors
Risk assets
Funds 1992-2019
−0.0001 (p = 0.990)
−0.0115 (p = 0.159)
+0.0011 (p = 0.888)
+0.0079 (p = 0.210)
−0.0209 (p = 0.004)
−0.030
−0.015
0.000
0.015
0.030
Mean FZ0 loss difference, richer minus baseline (negative: richer model better)